

RUNQI WANG

✉ allen-w@sjtu.edu.cn · ☎ (+86) 18800340885



🎓 EDUCATION

Shanghai Jiao Tong University

Sep. 2023 – Jun. 2027 (Expected)

B.S. in Mathematics and Applied Mathematics & Artificial Intelligence

Undergraduate, Rank : 2 / 37

Major Courses: Probability Theory, Mathematical Statistics, Stochastic Processes, Design and Analysis of Algorithms, Scientific Computing, Machine Learning, Deep Learning and Applications.

🧑‍🔬 RESEARCH AND PROJECT EXPERIENCE

Unlearning Offline Stochastic Multi-Armed Bandits

Jul. 2025 – Jan. 2026

Research

Co-first Author

Submitted to NeurIPS 2026 (CCF-A)

- Studied machine unlearning for offline stochastic multi-armed bandits, focusing on decision-making under data deletion, privacy constraints, and uncertainty.
- Designed an adaptive algorithm combining **Gaussian mechanism** and **rollback** based on sample size and privacy budget; contributed to regret-style theoretical analysis and upper/lower bound proofs.
- Built large-scale simulation pipelines and validated the theoretical trade-off, achieving **20%–70%** expected performance improvement over baseline methods across multiple settings.

Efficient Stochastic Combinatorial Semi-Bandits

Feb. 2025 – Jun. 2025

Research

Second Author

Submitted to ESA 2026 (CCF-B)

- Worked on stochastic combinatorial semi-bandit algorithms, with emphasis on efficient exploration, regret minimization, and scalable decision-making over large action spaces.
- Contributed to UCB-based algorithm design and theoretical analysis, deriving near-optimal regret guarantees under computationally efficient implementations.
- Led experiments on semi-bandit and cascading-bandit environments; results show over **40%** average improvement over baselines in sample efficiency and computational efficiency.

Spatiotemporal Optimization Based on MFHL-PSO

Sep. 2025

Competition

Team Leader; Modeling and Writing

CUMCM National 1st Prize (Top 0.5%)

- Led mathematical modeling, algorithm design, and paper writing; achieved national top **0.5%** performance with significantly reduced computational cost.
- Proposed a multi-fidelity hierarchical particle swarm optimization algorithm with adaptive search, constriction factor, and elite re-evaluation, improving convergence speed and optimization stability.

👤 LEADERSHIP EXPERIENCE

Student Union, School of Mathematical Sciences, SJTU

President

Dec. 2024 – Nov. 2025

💖 HONORS AND AWARDS

Mathematical Contest in Modeling, Finalist Prize (Top 1%)

2026

Merit Student / Outstanding Student Leader

2024 / 2025

Scholarship of Tung's Foundation / Guo Chenchen Scholarship, SJTU

2024 / 2025

⚙️ SKILLS

- Programming: Proficient in Python, C++ and MATLAB; experienced in deep learning model training (Top 1% of SJTU's Kaggle Competitions). Familiar with PyTorch, NumPy, Pandas, and Anaconda.
- Language: Fluent in Chinese (native), English; experienced in academic communication in English.